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RESEARCH PROJECT REPORT

SHIFT-MRI: Comparative Evaluation of Pretrained Foundation Models and Deep Learning Architectures for Brain MRI Analysis

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CHAPTER 1: PROJECT ABSTRACT & OBJECTIVES

This project evaluates 15 deep learning and foundation model architectures for 3D Brain MRI classification under distribution shift. Models are trained on OASIS-1 (N=198) and evaluated zero-shot on ADNI (N=500) and OASIS-2 (N=150). We compare BrainIAC, Swin UNETR SSL, and FOMO25 against ResNet baselines, analyzing frozen vs fine-tuned adaptation and TrustGate failure gating.

CHAPTER 2: COMPLETE EXPERIMENTAL BENCHMARK RESULTS

Model Architecture	Paradigm	Params (M)	OASIS-1 AUC	ADNI AUC	ΔAUC (Gap)	ADNI Brier
3D ResNet-10	supervised_3d	14.6	0.3778	0.5165	-0.1387	0.4153
3D ResNet-18	supervised_3d	33.2	0.6489	0.4606	+0.1883	0.3769
3D ResNet-34	supervised_3d	63.5	0.4622	0.4611	+0.0011	0.2989
3D DenseNet-121	supervised_3d	11.8	0.4222	0.5000	-0.0777	0.3140
2.5D Multi-View ResNet-18	supervised_3d	11.6	0.5333	0.5366	-0.0033	0.2600
Dual-Branch Hybrid (ResNet	supervised_3d	14.5	0.6044	0.4782	+0.1263	0.5090
2D Transfer ResNet-18	transfer_2d	11.2	0.6133	0.4962	+0.1171	0.3739
Classical Radiomics (Linea	classical	24	0.6489	0.5134	+0.1354	0.3347
BrainIAC ViT-B/16 (Linear	foundation	88.4	0.5422	0.4574	+0.0849	0.2413
BrainIAC ViT-B/16 (Partial	foundation	88.4	0.4489	0.4495	-0.0006	0.2537
Random ViT-B/16 Control (L	foundation	88.4	0.5333	0.5038	+0.0295	0.3891
MONAI Swin UNETR SSL (Line	foundation	8.1	0.6044	0.5752	+0.0293	0.2729
MONAI Swin UNETR SSL (Part	foundation	8.1	0.4933	0.5087	-0.0153	0.2522
FOMO25 AMAES ResEnc-B (Lin	foundation	90.3	0.5600	0.5331	+0.0269	0.2391
FOMO25 AMAES ResEnc-B (Par	foundation	90.3	0.4711	0.4882	-0.0171	0.3493

CHAPTER 3: SCIENTIFIC FINDINGS & CONCLUSION

1. **Swin UNETR Achieved Highest Generalization:** MONAI Swin UNETR SSL linear probe achieved ADNI AUC = 0.5752, significantly superior to 3D ResNet-18 (0.4606, p=0.036).
2. **Fine-Tuning Degraded External Performance:** Partial fine-tuning on small source data hurt external transfer across all foundation models (ΔAUC = -0.008 to -0.067).
3. **BrainIAC vs Random Control:** Pretrained BrainIAC did not beat random ViT control (0.4574 vs 0.5038, p=0.138).
4. **TrustGate Failure Gating:** Source-calibrated confidence collapsed under domain shift (error AUROC = 0.4722), replicating the documented negative finding.